

G. Navaneeth Singh

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ABOUT

Third-year CSE student focused on ML systems, GPU programming, and efficient LLM inference. I build systems from first principles to understand and optimize what happens below high-level frameworks, with work spanning Triton kernels, quantization, operator fusion, and native C++ inference. Seeking research opportunities in ML systems, GPU computing, and efficient inference.

EDUCATION

JSS Academy of Technical Education, Bengaluru

2024 – 2028

B.E. Computer Science & Engineering · VTU · 5th Semester **CGPA: 8.44**

TECHNICAL SKILLS

Languages: C++17/20, Python, C, Java, SQL

ML / GPU: PyTorch, OpenAI Triton, LibTorch, NF4 Quantization, Kernel / Operator Fusion, PyBind11, NumPy, scikit-learn

Systems: Clang/LLVM, GCC, CMake, QEMU, GDB, Valgrind, Emscripten / WebAssembly

Backend / Dev: FastAPI, MySQL, Git, GitHub, WSL2 / Ubuntu

PROJECTS

Hydra Engine — LLM Inference Infrastructure

[GitHub](#)

Python · C++17 · OpenAI Triton · PyBind11 · LibTorch

- Custom LLM inference framework for NVIDIA Ada Lovelace GPUs with fused Triton kernels, NF4 quantization, PyBind11/LibTorch integration, and a native C++ generation loop.
- Achieved 1.68× speedup and 68% fewer VRAM round-trips vs PyTorch 2.x baselines; runs 1.1B+ parameter models on 6 GB VRAM with zero Python in the hot path.

Leaf — CPU Neural Network Inference Runtime

[GitHub](#)

C++17 · AVX2 · ONNX · Python · CMake

- Built a CPU-native neural network inference runtime with a C++ graph executor, ONNX-to-Leaf IR pipeline, Conv/BatchNorm/ReLU fusion, live-activation reclamation, and AVX2 kernels.
- Matched PyTorch ResNet-18 logits within 3.93e-06 max absolute difference and reduced mean latency from 397.4 ms to 140.9 ms on the development WSL machine using an AVX2 4x8 GEMM micro-kernel.

LPG Distribution Management System

[GitHub](#)

FastAPI · Python · MySQL · JavaScript · Leaflet.js

- Built a FastAPI/MySQL platform centered on priority-based LPG allocation, with QR-based delivery verification, agency-proximity mapping, REST APIs, and role-based access control.

COMPETITIONS

AI Prompt Battle — Global Technical School

Offline · Team

- Competed across multiple rounds focused on precision prompting, model-behavior steering, and adversarial prompt design.

Embedded Systems Challenge — NIT Surathkal

Online · National

- Worked through national-level problem statements involving real-time constraints, hardware-software integration, and low-level system design.

LANGUAGES

English · Professional **Kannada** · Native **Hindi** · Conversational